

BLE STACK INITIALIZATION

Step-by-Step Procedure to Enable BLE Stack on STM32WB55CG

This document explains how to upgrade the **FUS (Firmware Upgrade Service)** and **BLE Wireless Stack** on the **STM32WB55CG** series microcontroller.

This procedure is intended for **beginners/freshers**.

1. Prerequisites

Before starting, ensure you have the following:

Hardware

- STM32WB55CG development board
- ST-LINK debugger (or onboard ST-LINK)
- USB cable

Software

- **STM32CubeProgrammer** (latest version installed on PC)
- BLE mobile application (e.g., **nRF Connect** app)

Firmware Files (.bin)

You must download **two binary files** from the **official STMicroelectronics website**:

1. **FUS firmware**
Example:
2. `stm32wb5x_FUS_fw.bin`
3. **BLE Stack firmware**
Example:
4. `stm32wb5x_BLE_Stack_full_fw.bin`

⚠ Important Note:

The `.bin` files are **device-specific**.

- For STM32WB55CG → WB5x series files
- For other devices like **WB55RG, WB55HG, WBA series**, the firmware files will be different.

Always download the correct `.bin` files for your controller from the **ST official website**.

2. Hardware Connection

1. Connect the STM32WB55CG board to the **ST-LINK** (or use onboard ST-LINK).
 2. Connect the ST-LINK to your PC using a USB cable.
 3. Power ON the board.
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3. Open STM32CubeProgrammer

1. Launch **STM32CubeProgrammer**.
2. Select the **ST-LINK** interface.
3. Click **Connect**.

After successful connection:

- The ST-LINK and target board will be detected.
 - The **device information and memory address** of the controller will be displayed.
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4. Upgrade the FUS Firmware

1. In STM32CubeProgrammer, go to the **FUS** section (represented by a **green Wi-Fi-like icon** at the top-left corner).
2. Click **Start Address Automatic Determination**.
3. Click **Browse** and select the **FUS firmware (.bin)** file.
4. Click **Firmware Upgrade**.

During this step:

- A pop-up window will show the detected firmware information (for example: *FUS version compatible with Stack v1.23.0*).
- Confirm and proceed with the upgrade.

Wait until the FUS upgrade completes successfully.

5. Upgrade the BLE Stack Firmware

1. In the same FUS section, click **Browse** again.
 2. Select the **BLE Stack firmware (.bin)** file.
 3. Click **Firmware Upgrade**.
 4. Wait until the BLE stack flashing process completes.
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6. Verify FUS and BLE Stack Versions

1. Click **Read FUS Info**.

The following information will be displayed:

- **FUS Version:** e.g., v2.1.0.0
- **BLE Stack Version:** e.g., v1.23.0.3
- **FUS Status:** FUS_NO_ERROR

This confirms that both the FUS and BLE stack are successfully installed.

7. Start the Wireless Stack

1. Click **Start Wireless Stack**.
 2. This action activates the BLE wireless stack on the controller.
 3. Wait for the process to complete.
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8. Disconnect the Board Safely

1. Click the **Disconnect** button in STM32CubeProgrammer.
2. Only after disconnecting, unplug the USB cable.

⚠ **Do not remove the cable without clicking Disconnect**, as it may cause communication issues.

9. BLE Application Verification

1. Flash a **BLE example application** (using STM32CubeIDE or any preferred IDE).
2. Power the board.
3. Open the **nRF Connect** BLE app on your mobile phone.
4. Scan for nearby BLE devices.

5. Verify that your STM32WB55CG BLE device appears in the list.

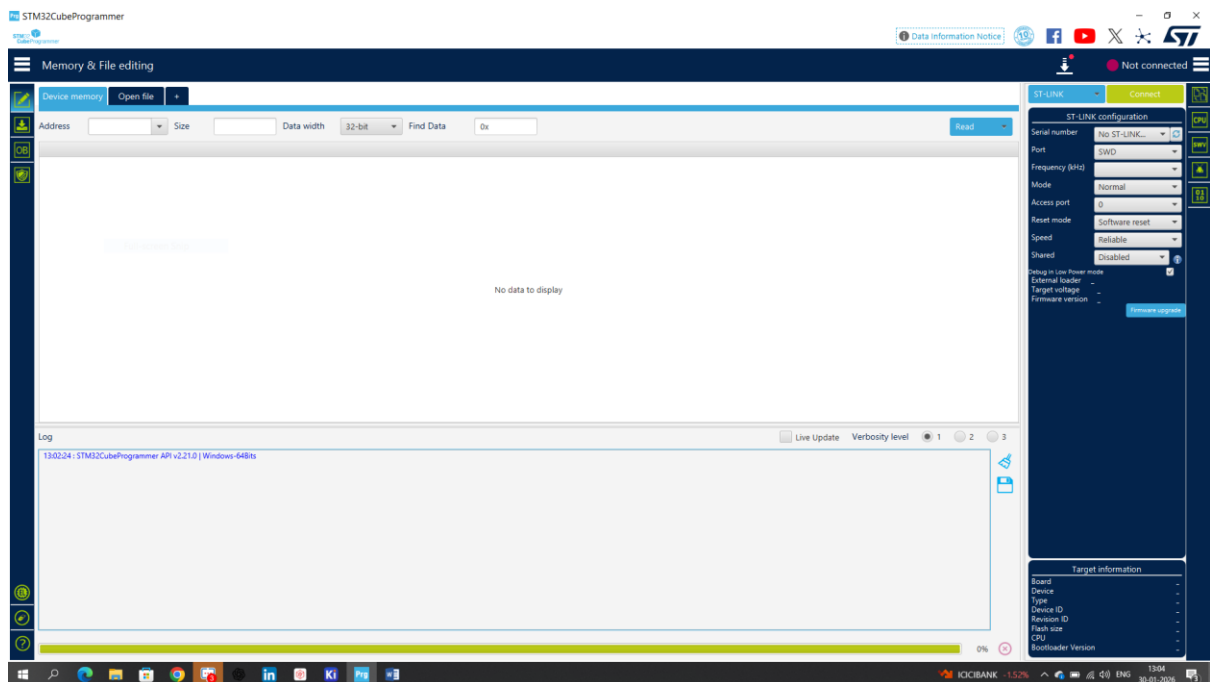
10. Conclusion

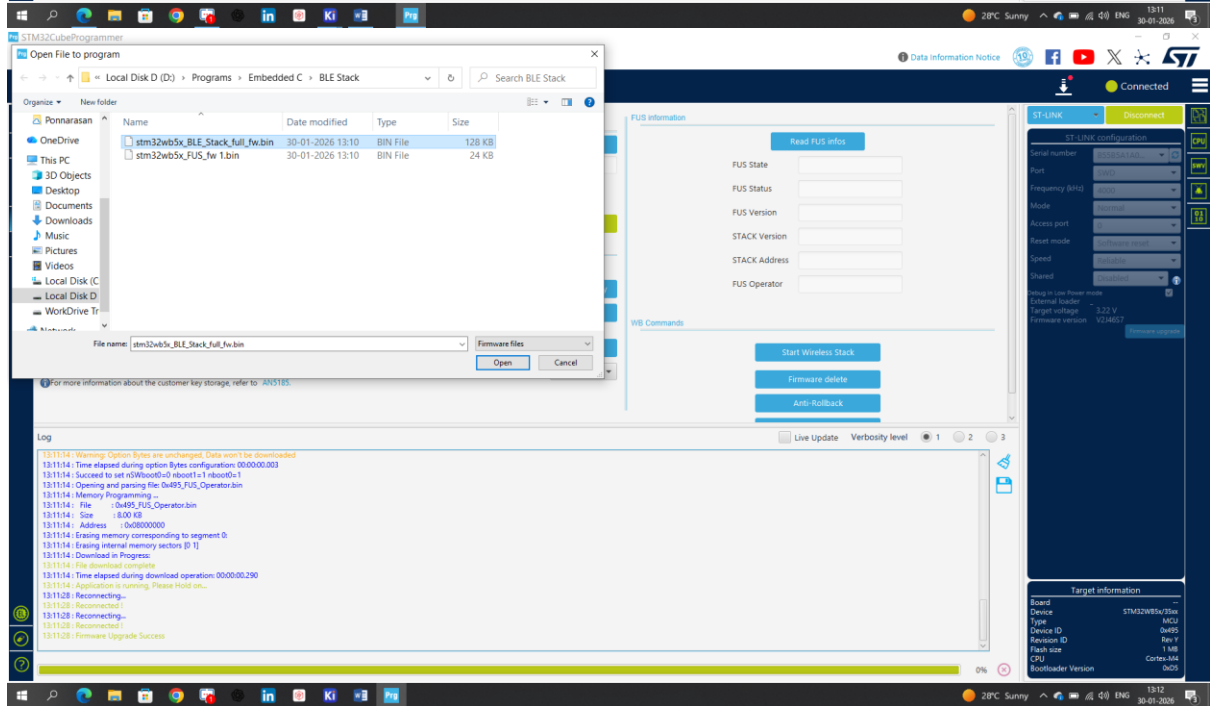
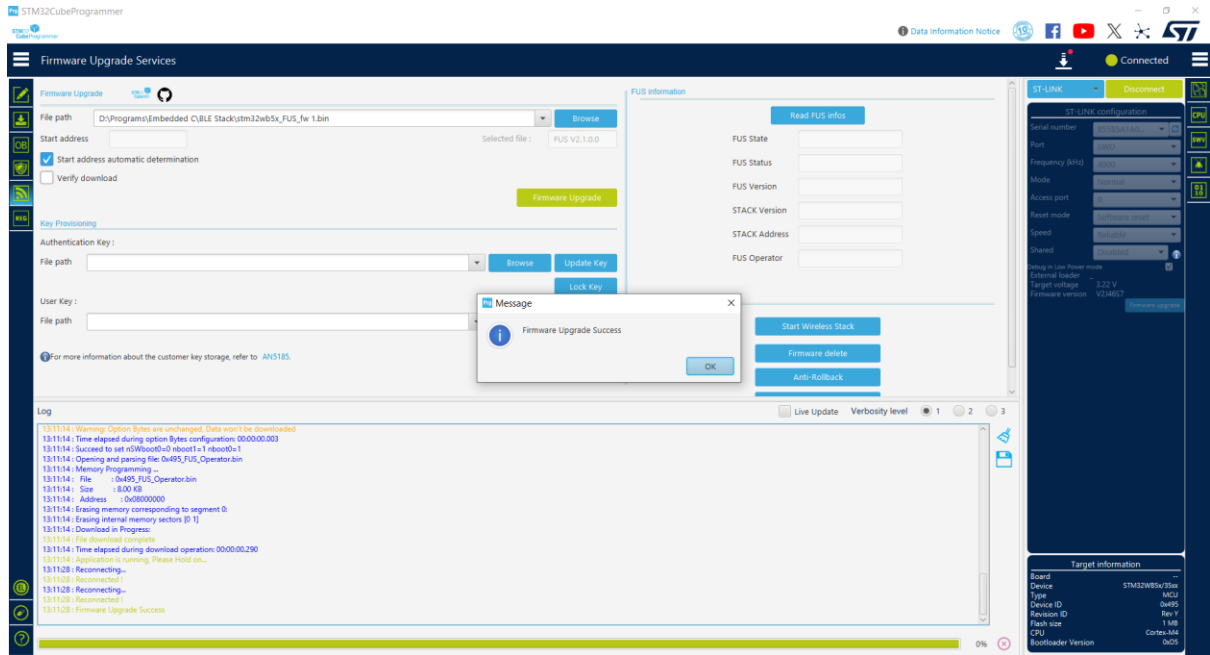
You have successfully:

- Upgraded the **FUS firmware**
- Installed the **BLE Wireless Stack**
- Verified BLE operation using a mobile application

This procedure is essential before running any BLE application on STM32WB series microcontrollers.

Reference Pictures:





STM32CubeProgrammer

Data Information Notice

ST

Firmware Upgrade Services

Firmware Upgrade

File path:

Start address: Selected file: STACK V1.23.0

☒ Start address automatic determination

☐ Verify download

Key Provisioning

Authentication Key:

File path:

User Key:

File path:

For more information about the customer key storage, refer to [AN5185](#).

FUS Information

FUS State:

FUS Status:

FUS Version:

STACK Version:

STACK Address:

FUS Operator:

ST-LINK

ST-LINK configurations

Serial number:

Port:

Frequency (kHz):

Mode:

Access port:

Reset mode:

Speed:

Shared:

Using in Live Press mode

Current loader:

Target voltage: 3.22 V

Firmware version: V2.06.07

Target information

Board:

Device: STM32WB5x/35x

Type: MCU

Device ID: 0x495

Revision ID: Rev 1

Flash size: 1 MB

CPU: Cortex-M4

Bootloader Version: 0x05

Log

☐ Live Update Verbose level: 1 2 3

13:13:49 : Warning: Option Byte nboot1, value 0x1, was not modified.

13:13:49 : Warning: Option Bytes are unchanged. Data won't be downloaded

13:13:49 : Time elapsed during option Bytes configuration: 00:00:00.000

13:13:49 : Succeeded to set nRbboot0=0 nboot1=1 nboot0=1

13:13:49 : Opening and parsing file 0x495_FUS_Operator.bin

13:13:49 : Memory Programming ...

13:13:49 : File : 0x495_FUS_Operator.bin

13:13:49 : Size : 1300 B

13:13:49 : Address : 0x08000000

13:13:49 : Erasing memory corresponding to segment 0

13:13:49 : Erasing internal memory sectors [0..1]

13:13:49 : Download in Progress

13:13:49 : File download complete

13:13:49 : Time elapsed during download operation: 00:00:00.299

13:13:49 : Application is running. Please Hold on...

13:13:50 : Reconnecting...

13:13:50 : Reconnected !

13:13:51 : FocStarNWS activated successfully

13:13:51 : Start Wireless Stack command execution finished with success

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